

RISC-V RV32I Base Instruction Set

[1] RISC-V RV32I Instruction Set Listing, Table 25.2, page 130

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vicilogic RV32I Base Instruction Set

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Interactive RISC-V RV32I Base Instruction Set Table

ID (0x)	ID (0d)	instruction mnemonic	description	instruction(31:0) format							Type	instruction syntax	
31121176opcode0													
1	1	LUI	load upper immediate	imm[31:12]			rd	0110111			U	lui rd, imm	
2	2	AUIPC	add upper immed to PC	imm[31:12]			rd	0010111			U	auipc rd, imm	
Jump and link													
3	3	JAL	jump and link	imm[20;10;11;19:12]			rd	1101111			J	jai rd, imm	
Branch instructions													
312524201915141211760 f3													
4	4	BEQ	branch if equal	imm[12;10;5]	rs2	rs1	000	imm[4;1;11]	1100011	B	beq rs1, rs2, imm		
5	5	BNE	branch if not equal	imm[12;10;5]	rs2	rs1	001	imm[4;1;11]	1100011	B	bne rs1, rs2, imm		
6	6	BLT	branch if less than (lt)	imm[12;10;5]	rs2	rs1	100	imm[4;1;11]	1100011	B	blt rs1, rs2, imm		
7	7	BGE	branch if greater than (gt)	imm[12;10;5]	rs2	rs1	101	imm[4;1;11]	1100011	B	bge rs1, rs2, imm		
8	8	BLTU	branch if lt than or eq unsgn	imm[12;10;5]	rs2	rs1	110	imm[4;1;11]	1100011	B	bltu rs1, rs2, imm		
9	9	BGEU	branch if gt than unsgn	imm[12;10;5]	rs2	rs1	111	imm[4;1;11]	1100011	B	bgeu rs1, rs2, imm		
Jump and link (registered)													
0xA	10	JALR	jump & link register	imm[11:0]			rs1	000	rd	1100111	I	jair rd, imm(rs1)	
Memory Load Instructions													
0xB	11	LB	load byte	imm[11:0]			rs1	000	rd	0000011	I	lb rd, imm(rs1)	
0xC	12	LH	load halfword	imm[11:0]			rs1	001	rd	0000011	I	lh rd, imm(rs1)	
0xD	13	LW	load word	imm[11:0]			rs1	010	rd	0000011	I	lw rd, imm(rs1)	
0xE	14	LBU	load byte unsigned	imm[11:0]			rs1	100	rd	0000011	I	lbu rd, imm(rs1)	
0xF	15	LHU	load halfword unsigned	imm[11:0]			rs1	101	rd	0000011	I	lhu rd, imm(rs1)	
Integer register-immediate instructions													
0x10	16	ADDI	add immediate	imm[11:0]			rs1	000	rd	0010011	I	addi rd, rs1, imm	
0x11	17	SLTI	set less than immediate	imm[11:0]			rs1	010	rd	0010011	I	slti rd, rs1, imm	
0x12	18	SLTIU	set less than immed unsgn	imm[11:0]			rs1	011	rd	0010011	I	sltiu rd, rs1, imm	
0x13	19	XORI	xor immediate	imm[11:0]			rs1	100	rd	0010011	I	xori rd, rs1, imm	
0x14	20	ORI	or immediate	imm[11:0]			rs1	110	rd	0010011	I	ori rd, rs1, imm	
0x15	21	ANDI	and immediate	imm[11:0]			rs1	111	rd	0010011	I	andi rd, rs1, imm	
Memory Store Instructions													
0x16	22	SB	store byte	imm[11:5]			rs2	rs1	000	imm[4:0]	0100011	S	sb rs2, imm(rs1)
0x17	23	SH	store halfword	imm[11:5]			rs2	rs1	001	imm[4:0]	0100011	S	sh rs2, imm(rs1)
0x18	24	SW	store word	imm[11:5]			rs2	rs1	010	imm[4:0]	0100011	S	sw rs2, imm(rs1)
Constant shift Instructions													
f7													
0x19	25	SLLI	constant-shift left	0000000			shamt	rs1	001	rd	0010011	I	slli rd, rs1, shamt
0x1A	26	SRLI	constant-shift right	0000000			shamt	rs1	101	rd	0010011	I	srl rd, rs1, shamt
0x1B	27	SRAI	constant-shift right arithmetic	0100000			shamt	rs1	101	rd	0010011	I	srai rd, rs1, shamt
Integer register-register instructions													
0x1C	28	ADD	add	0000000			rs2	rs1	000	rd	0110011	R	add rd, rs1, rs2
0x1D	29	SUB	subtract	0100000			rs2	rs1	000	rd	0110011	R	sub rd, rs1, rs2
0x1E	30	SLL	register-shift left	0000000			rs2	rs1	001	rd	0110011	R	sll rd, rs1, rs2
0x1F	31	SLT	set less than	0000000			rs2	rs1	010	rd	0110011	R	slt rd, rs1, rs2
0x20	32	SLTU	set less than unsigned	0000000			rs2	rs1	011	rd	0110011	R	sltu rd, rs1, rs2
0x21	33	XOR	xor	0000000			rs2	rs1	100	rd	0110011	R	xor rd, rs1, rs2
0x22	34	SRL	register-shift right (logical)	0000000			rs2	rs1	101	rd	0110011	R	srl rd, rs1, rs2
0x23	35	SRA	register-shift right (arithmetic)	0100000			rs2	rs1	101	rd	0110011	R	sra rd, rs1, rs2
0x24	36	OR	or	0000000			rs2	rs1	110	rd	0110011	R	or rd, rs1, rs2
0x25	37	AND	and	0000000			rs2	rs1	111	rd	0110011	R	and rd, rs1, rs2
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0x26	38	FENCE		fm	pred	succ	rs1	000	rd	0001111			
0x27	39	ECALL		000000000000			00000	000	00000	1110011			
0x28	40	EBREAK		000000000001			00000	000	00000	1110011			